

Intervention matrix $\mathbf{B}$

ecoinvent 3.8 cut-off

direct flow intensities $b_{e,p}$

$\times$

Scaling vector $\dot{\mathbf{s}}$

annual production volumes $\dot{s}_p = PV_p$

sources

ecoinvent · trade statistics ·
reports · literature

$=$

Normalization inventory $\dot{\mathbf{g}}$

$$\dot{\mathbf{g}} = \mathbf{B} \dot{\mathbf{s}}$$

global annual amount of each flow

$\times \mathbf{Q}$ characterization matrix

(EF v3.0, ReCiPe 2016, IMPACTWorld+)



Normalization reference $\dot{\mathbf{h}}$

$$\dot{\mathbf{h}} = \mathbf{Q} \dot{\mathbf{g}}$$

per impact category i

Applied to a product
system under study:

$$\tilde{h}_i = h_i / \dot{h}_i \quad \text{(unit: yr)}$$